REVIEW – 1

Veridex

A Rule-Based Public Information Reliability & Verification System

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Scope of the Project

The scope of Veridex encompasses the design, development, and implementation of a full-stack, web-based information reliability platform that facilitates structured assessment of publicly circulating claims. The system integrates authenticated user participation, deterministic rule-based reliability evaluation, AI-assisted advisory analysis, and centralized moderation within a unified architectural framework.

The platform is designed to provide a transparent and auditable mechanism for analysing public claims without issuing binary truth judgments. Instead of categorizing information as “true” or “false,” the system generates structured reliability indicators derived from predefined, explainable rules.

The project scope includes the following functional and technical boundaries:

- Enabling authenticated and registered users to submit publicly shared claims in a structured format.
- Allowing users to attach supporting references, including URLs and uploaded documents, to strengthen contextual evaluation.
- Implementing a deterministic rule-based evaluation engine that computes reliability confidence indicators based on predefined measurable criteria.
- Integrating AI-assisted structured claim analysis for informational and advisory insights without granting AI authority over reliability outcomes.
- Providing claim-specific, moderated discussion threads to encourage structured community engagement.
- Implementing secure login and signup functionality with role-based authorization mechanisms.
- Granting moderator-level access to oversee, review, and manage claims, discussions, and uploaded sources.
- Ensuring transparency, neutrality, and auditability of all system processes through explainable architectural design.

The system explicitly excludes:

- Automated declaration of claims as absolutely true or false.
- Replacement of professional fact-checking or journalistic institutions.
- Suppression of dissenting views without human moderation oversight.
- Black-box AI decision-making that lacks explainability.

Thus, Veridex operates as a structured reliability indicator and moderated discussion platform rather than a verdict-delivering fact adjudication system.

Objectives

The primary objectives of Veridex are designed to address the technical, structural, and governance challenges associated with misinformation analysis in digital ecosystems.

The major objectives include:

1. To design and implement a secure authentication system incorporating login and signup functionality with encrypted password storage.
2. To establish a robust role-based access control mechanism differentiating between standard Users and Moderators.
3. To develop a structured claim submission system that supports contextual evidence attachment.
4. To design and implement a deterministic rule-based reliability evaluation engine that operates independently of AI verdict authority.
5. To integrate AI-assisted structural claim analysis strictly for advisory and organizational purposes.
6. To enable claim-specific threaded discussions with voting mechanisms to promote structured user engagement.
7. To implement a centralized moderation dashboard allowing oversight of platform activity and governance actions.
8. To ensure system transparency, explainability, and auditability through modular architecture and rule documentation.
9. To maintain data integrity, controlled access, and secure file handling across all system components.
10. To design a scalable backend architecture capable of supporting future enhancements and increased user participation.

Proposed System Architecture

Veridex follows a modular, layered full-stack architecture based on the Model–View–Controller (MVC) design pattern combined with role-based access control principles. The architecture is structured to ensure separation of concerns, maintainability, scalability, and security.

A. Presentation Layer (Frontend)

The Presentation Layer is developed using HTML, CSS, and Vanilla JavaScript. It serves as the user interaction interface and communicates with backend services via RESTful APIs using asynchronous HTTP requests.

The frontend provides:

- User authentication interfaces (Login and Signup forms).
- Structured claim submission interface with source attachment capability.
- Dynamic claim listing and status display.
- Claim-specific discussion interface with voting and threaded replies.
- Restricted moderator dashboard accessible only to authorized users.

Authentication tokens (such as JSON Web Tokens – JWT) are utilized to maintain secure user sessions and protect restricted routes.

The frontend is designed to dynamically render content based on user role and backend responses, ensuring responsive and secure interaction.

B. Application Layer (Backend – Express.js)

The Application Layer is implemented using Node.js and Express.js. It serves as the core processing engine of the platform and is structured into modular components.

1. Authentication Module

Responsible for:

- User registration and account creation.

- Secure password hashing using industry-standard cryptographic methods (e.g., bcrypt).
- User login verification.
- Token generation and validation.
- Implementation of role-based authorization middleware.

This module ensures secure identity management and controlled access to system resources.

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2. Claim Management Module

Responsible for:

- Creating and retrieving claims.
- Maintaining claim processing states.
- Updating claim status based on evaluation outcomes.
- Storing AI-assisted structured outputs associated with each claim.

This module ensures structured storage and lifecycle management of claim entities.

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3. Source Management Module

Responsible for:

- Handling URL-based source submissions.
- Managing secure file uploads.
- Categorizing source types (e.g., official, media, document).
- Associating sources with specific claims.

It ensures proper evidence linkage and secure file handling.

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4. Evaluation Engine Module

This module implements deterministic reliability assessment logic. It evaluates claims using predefined rules such as:

- Number of attached sources.
- Presence of official or authoritative sources.
- Temporal freshness of the claim.

The module generates reliability confidence indicators based solely on explainable rule conditions. The evaluation engine is separated from AI advisory components to preserve transparency.

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5. Discussion Module

This module manages claim-specific discussions and supports:

- Text-based posts.
- Image-based posts.
- Poll-based interactions.
- Upvote and downvote mechanisms.
- Threaded reply to structures.

All discussion activities are associated with authenticated users to ensure accountability and traceability.

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6. Moderator Module

The Moderator Module provides centralized oversight and governance capabilities. It enables moderators to:

- View all claims and discussions.
- Monitor user activity.
- Review flagged or suspicious content.
- Manage system-level actions and maintain platform integrity.

This module strengthens governance and platform accountability.

C. Data Layer (MongoDB)

The Data Layer uses MongoDB as a NoSQL database to store structured and semi-structured data efficiently.

Collections include:

- Users
- Claims
- Sources
- Discussions
- Verifications

Each entity is logically separated to ensure modularity, maintainability, and scalability. Relationships between collections are managed using references and identifiers.

D. Security Layer

The Security Layer enforces:

- Secure password hashing (e.g., bcrypt).
- Token-based authentication using JWT.
- Role-based route protection middleware.
- Input validation and sanitization.
- Controlled file upload handling.
- Protection against unauthorized data access.

This layer ensures confidentiality, integrity, and secure session management.

List of Modules

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1. User Authentication Module

This module handles secure user registration and login processes. It encrypts passwords using cryptographic hashing algorithms, assigns appropriate user roles (User or Moderator), generates authentication tokens, and manages session validation to protect restricted resources.

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2. Claim Submission Module

This module enables authenticated users to submit publicly circulating claims. It stores claim text, timestamps, and processing status while ensuring structured data representation within the database.

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3. Source Attachment Module

This module allows users to attach references in the form of URLs or uploaded documents. It categorizes sources based on predefined classifications and links them to corresponding claims for evaluation.

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4. Rule-Based Evaluation Module

This module applies deterministic, explainable rules to generate reliability confidence indicators. It ensures transparency by basing evaluations solely on measurable parameters rather than probabilistic black-box models.

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5. AI Advisory Analysis Module

This module performs structural claim breakdown, detects similar claims, identifies potential conflicts, and generates informational summaries. AI outputs are strictly advisory and do not influence the deterministic reliability scoring mechanism.

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6. Discussion Module

This module facilitates structured discussion under each claim. It supports voting, threaded replies, and multimedia posts while ensuring user accountability through authenticated participation.

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7. Moderator Control Module

This module provides centralized administrative access for reviewing platform activity. It enables monitoring of claims, discussions, and user actions to maintain platform governance and integrity.

Investigation

S. No.	Patent / Publication	Description	Relevance to Proposed Invention	Identified Gap
1	US Patent – Automated Fact Verification Systems	Describes AI-based systems that automatically classify claims as true or false using machine learning models.	Shows automated misinformation detection techniques using probabilistic classification.	Black-box decision making; lacks explainability and structured moderation workflow.
2	IEEE Publication – Fake News Detection using Machine Learning	Uses NLP and supervised learning algorithms to classify news content into fake or real categories.	Demonstrates automated classification frameworks for misinformation detection.	Binary verdict approach; no lifecycle modeling; no user governance integration.
3	ACM Research – Credibility Assessment in Online Social Networks	Analyzes user credibility using network propagation models and source reputation scoring.	Highlights importance of source credibility and network influence in reliability evaluation.	Relies heavily on social metrics; lacks deterministic rule-based transparency.
4	US Patent – Content Moderation Framework with Multi-Level Review	Describes layered moderation systems involving automated detection and human review stages.	Like moderator review layer in Veridex, supports governance structure.	Does not integrate structured reliability evaluation logic tied to claims.

Summary

Veridex is a structured claim reliability and moderated discussion platform designed to address the limitations of existing factchecking and misinformation detection systems.

Unlike traditional binary classification systems that label content as “true” or “false,” Veridex provides deterministic reliability indicators based on transparent rule-based evaluation parameters. The system integrates authenticated user participation, role-based governance (User, Verified User, Moderator), structured discussions, and a defined claim lifecycle model.

The platform bridges the gap between automated evaluation systems and community-driven discussion frameworks by combining explainable rule logic with moderated engagement.

Working Principle

The working principle of Veridex is based on a layered, rule-driven workflow:

Step 1: Claim Submission

A registered user submits a claim along with one or more supporting sources.

Step 2: Input Validation

The system validates claim format, completeness, and user eligibility.

Step 3: Source Attachment

Users attach references (URLs or documents). Multiple sources may be added.

Step 4: Deterministic Evaluation

The Evaluation Module performs parallel checks:

- Source count validation
- Official source detection
- Freshness verification

These checks operate concurrently (fork-join structure) and feed into a rule-based evaluation engine.

Step 5: Confidence Computation

Based on predefined deterministic rules, the system calculates a confidence level (High / Medium / Low).

Step 6: Reliability Categorization

The claim is categorized as:

- Reliable
- Partially Reliable
- Low Confidence

Step 7: Moderation Workflow

If flagged, the claim undergoes moderator review, which may result in:

- Approval
- Rejection
- Re-evaluation

Step 8: Finalization

Approved claims may be archived. Rejected or discarded claims enter final states.

The system ensures that AI-assisted structural analysis is advisory and does not influence the deterministic reliability score.

Experimental Results

Since Veridex is a system architecture project, experimental validation is conceptual and design-oriented rather than dataset-driven.

Observed Outcomes During System Testing:

- Claims with multiple verified sources resulted in higher confidence classification.
- Claims lacking sufficient sources were categorized as low confidence.
- Parallel evaluation reduced logical processing delay.
- Role-based moderation improved governance and reduced misuse.

- Self-loop source addition enabled iterative reliability improvement.

Performance Observations:

- Deterministic rule evaluation ensures consistent outputs for identical input conditions.
- The modular architecture improves maintainability and scalability.
- Fork-join evaluation structure enhances logical separation of checks.

The system demonstrates stable state transitions across draft, evaluation, moderation, and archival stages.

Why It Is Better Than Existing Systems

Veridex improves upon existing systems in the following ways:

1. Explainability

Unlike black-box AI models, Veridex uses deterministic rules, making reliability decisions transparent and auditable.

2. Structured Lifecycle Modelling

The system models a claim's lifecycle from draft to archival, including re-evaluation and moderation stages, which many systems lack.

3. Separation of AI and Decision Authority

AI outputs are advisory only and do not determine reliability status, reducing ethical concerns.

4. Integrated Governance

Role-based access control introduces structured accountability and moderation oversight.

5. Community + Evaluation Integration

Combines structured discussion mechanisms with formal reliability scoring.

6. Iterative Improvement Mechanism

Users can add sources post-evaluation, enabling dynamic re-evaluation and progressive confidence enhancement.

7. Multi-Level Review System

Includes both automated rule evaluation and moderator oversight.